INNOMATICS
RESEARCH LABS

INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

EazyDiner Restaurant Analysis

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About me

Background

- I am a **3rd-year B.Tech student** pursuing Computer Science Engineering.

Why you want to learn Data Science

- I want to learn Data Science to develop strong skills in data analysis, visualization, and problem-solving using real-world data. As a computer science student, I am interested in data-driven decision-making and aim to pursue a career in data-related roles.

Any work experience

- Hands-on experience with EDA, data preprocessing, and deriving insights through visualization

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Business Problem and Use case domain understanding

Domain

- Restaurant / Food Services Industry

Problem Statement

- The restaurant industry is highly competitive, and customers select restaurants based on location, cuisine, cost, and ratings.
- There is a need to analyze restaurant data to understand pricing, ratings, and city-wise distribution to support better decision-making.

Use Case

- Analyze restaurant data from the EazyDiner platform.
- Identify pricing patterns, rating trends, and restaurant distribution across cities.
- Understand customer preferences and market trends in the restaurant domain.

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Objective of the Project

- To collect restaurant data from the EazyDiner platform using web scraping
- To clean and preprocess the collected data for analysis
- To perform Exploratory Data Analysis (EDA) on restaurant features
- To analyze pricing, ratings, cuisines, and city-wise distribution
- To identify patterns, trends, and customer preferences across different cities

Web Scraping – Details

- **Website Used:** EazyDiner
- **Domain:** Restaurant / Food Services
- **Tools & Libraries:** Python, Requests, BeautifulSoup, Regular Expressions
- **Process Followed:**

  - Identified relevant restaurant listing pages
  - Extracted details such as restaurant name, city, cuisine, cost, and ratings
  - Scraped data across multiple pages for better coverage
  - Stored the extracted data into a structured format (DataFrame / CSV)

Data Overview

Total Records: 1077 Restaurants

Total Features: 8

Features:

- Restaurant Name
- City
- Location
- Cuisine
- Rating
- Cost for Two
- Restaurant URL
- Reviews / Votes (if available)

Data Cleaning Steps

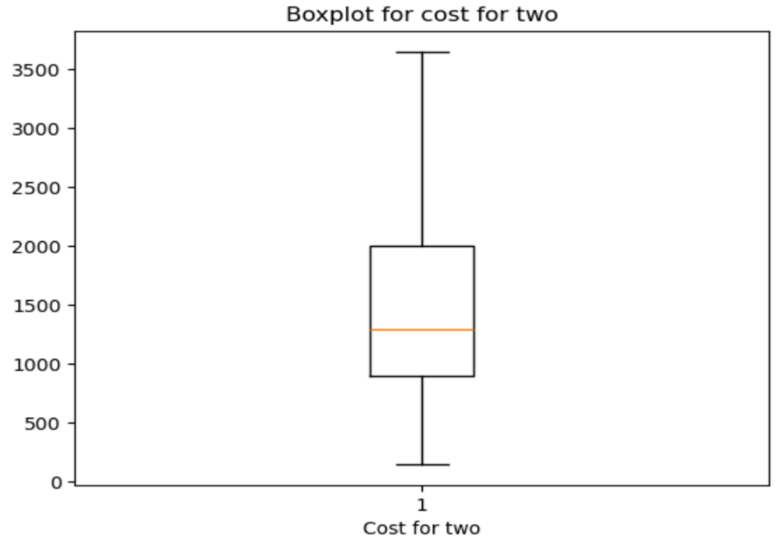
- Removed duplicate records and handled missing values
- Cleaned text data and standardized city & cuisine names
- Converted columns to appropriate data types
- Cleaned and validated cost-for-two values and finalized dataset for EDA

Data Manipulation Steps

- Created and renamed columns to improve data clarity and consistency
- Filtered the dataset based on city, rating, and cost criteria
- Performed city-wise and cuisine-wise grouping for analysis
- Applied aggregations and pivot tables to summarize key insights

Univariate Analysis - Numerical

Boxplot of cost for two:

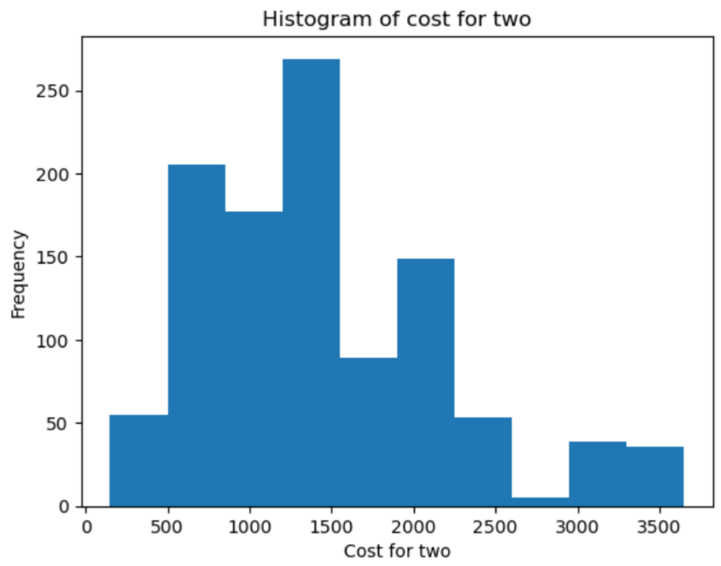


- The median cost for two is ~₹1,250, with most restaurants priced between ₹900 and ₹2,000.
- A few high-end restaurants (₹3,600+) create a right-skewed distribution, while budget options start

around ₹150–₹200.

Univariate Analysis - Numerical

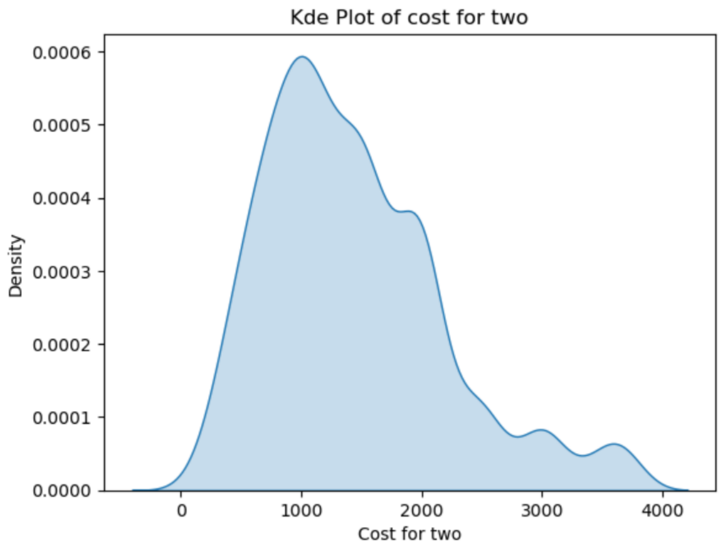
Histogram of Cost for two:



- Most restaurants fall in the ₹500–₹1,500 range, indicating a dominance of affordable to mid-range dining options.
- The frequency decreases at higher price ranges, with only a few premium restaurants priced above ₹2,500.

Univariate Analysis - Numerical

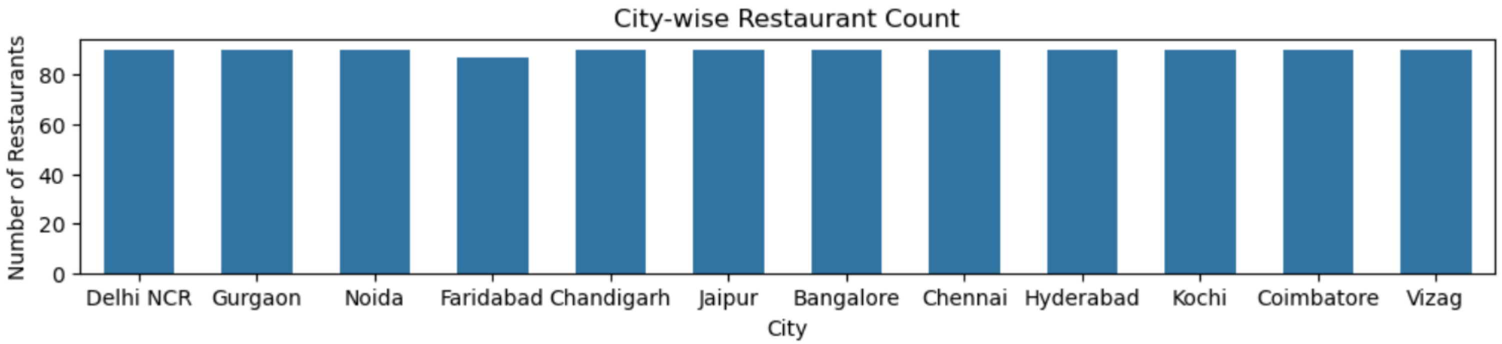
Kde plot for Cost for two:



- The highest density occurs around ₹800–₹1,200, indicating this is the most common price range for restaurants.
- Density gradually decreases as the cost increases, showing fewer restaurants in higher price ranges, with limited premium options above ₹2,500–₹3,000.

Univariate Analysis - Categorical

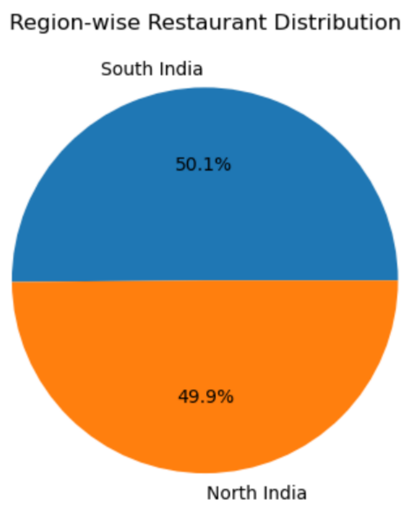
Count Plot of City-wise Restaurants:



- Restaurants are fairly evenly distributed across cities, indicating a balanced representation in the dataset.
- Faridabad shows a slightly lower restaurant count compared to other cities, but no city strongly dominates the distribution.

Univariate Analysis - Categorical

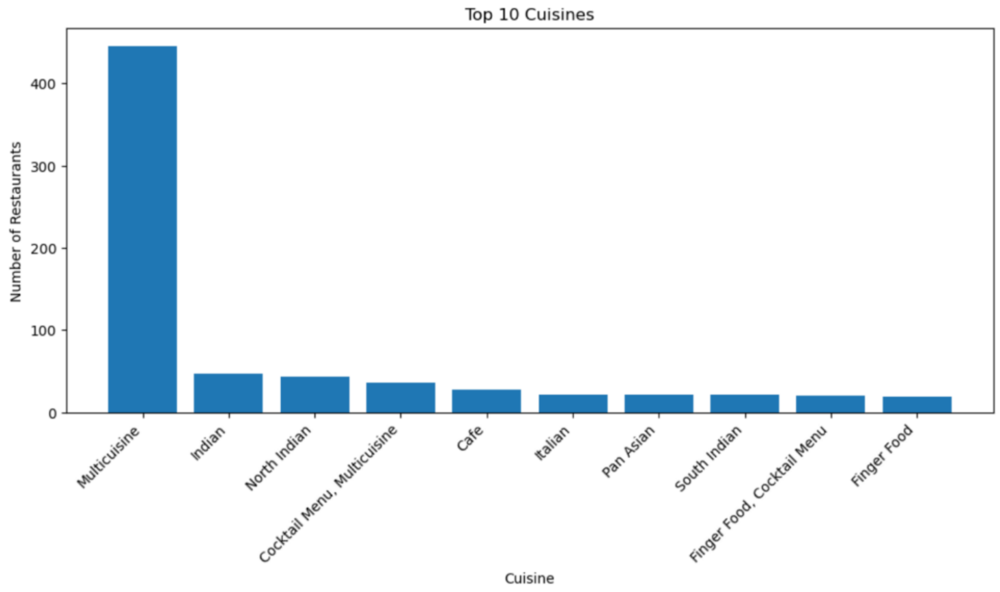
Region-wise Distribution of Restaurants :



- Restaurants are almost equally distributed between North and South India, with South India contributing ~50.1% and North India ~49.9%.
- This indicates a well-balanced regional representation in the dataset, with no significant regional bias.

Univariate Analysis - Categorical

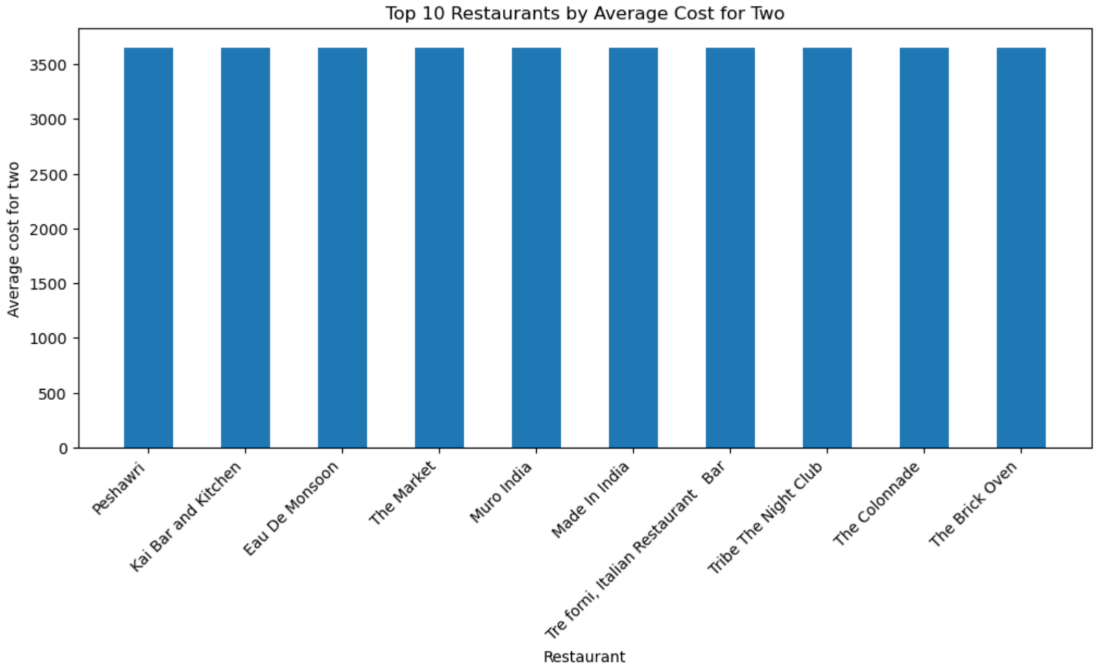
Cuisine-wise Distribution of Top 10 Restaurants :



- Multicuisine restaurants dominate the dataset, with a significantly higher count than any other cuisine type.
- Indian and North Indian cuisines follow at a much lower frequency, while other cuisines (Cafe, Italian, Pan Asian, South Indian, etc.) have relatively limited representation.

Univariate Analysis - Categorical

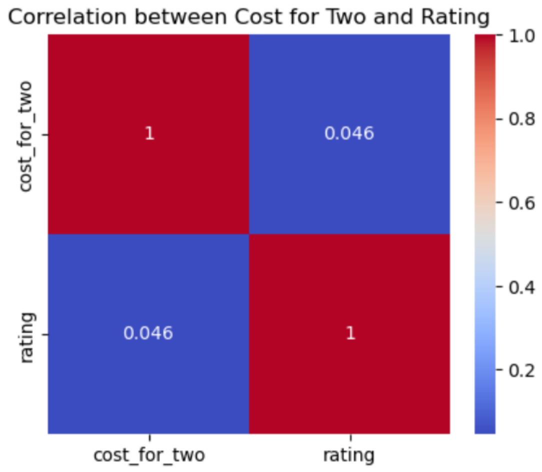
Premium Restaurants Based on Average Dining Cost



- The top 10 restaurants all fall in a similar high-price range, with an average cost for two around ₹3,600 – ₹3,700, indicating a premium dining segment.
- The minimal variation in bar heights shows that these restaurants are priced competitively within the luxury category, rather than having large cost differences.

Bivariate Analysis

Correlation Between Cost for Two and Restaurant Ratings :



- The correlation between cost for two and rating is very weak (~0.046), indicating no significant relationship between restaurant pricing and customer ratings.
- This suggests that higher-priced restaurants do not necessarily receive better ratings, and affordable restaurants can be rated just as highly.

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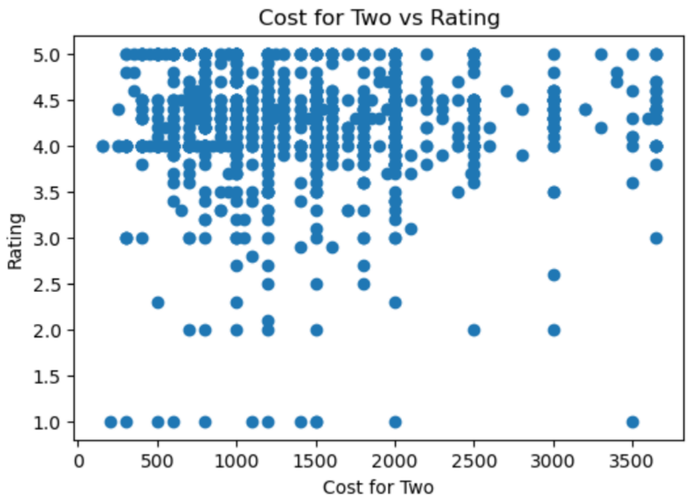
Bivariate Analysis

Cost for Two vs Restaurant Rating :



• Restaurant ratings are widely spread across all cost ranges, showing no clear upward or downward trend with price.

• Both low-cost and high-cost restaurants receive similar ratings, indicating that price does not strongly influence customer ratings.

Key Business Question

1. How do restaurant ratings vary across different cities?

- Restaurant ratings are fairly consistent across cities, with most ratings clustering between 4.0 and 4.5.
- No city shows a significantly higher or lower average rating, indicating uniform service quality across locations.

2. Which cuisines are most popular across cities?

- Multicuisine is the most popular cuisine across cities, followed by Indian and North Indian.

3. What is the relationship between cost for two and restaurant ratings?

- The correlation between cost and rating is very weak (~0.05).
- Higher prices do not guarantee better ratings, showing price has minimal impact on customer satisfaction.

4. Which cities have a higher concentration of restaurants?

- Restaurants are evenly distributed across cities, with no city showing strong dominance.

5. What factors most influence customer restaurant preferences?

- Quality and overall dining experience (ratings) influence customer preference more than price.

Conclusion

- Restaurants are evenly distributed across cities and regions, indicating a balanced dataset.
- Most restaurants fall in the affordable to mid-range cost segment, with fewer premium options.
- Multicuisine and Indian cuisines are the most popular across locations.
- There is no strong relationship between cost and ratings, showing that higher prices do not guarantee better customer satisfaction.
- Food quality and overall dining experience influence customer preferences more than price.

Experiences and Challenges

Experiences

- Gained practical exposure to data scraping, cleaning, and exploratory data analysis.
- Developed proficiency in visualizing and interpreting data to derive insights.
- Enhanced understanding of customer behavior and market trends.

Challenges

- Addressing missing and inconsistent data during preprocessing.
- Managing high-cardinality categorical variables such as cities and cuisines.
- Interpreting results where strong correlations were not observed.

Overall Learning

- Acquired hands-on experience in web scraping and real-world data preparation.
- Strengthened skills in exploratory data analysis and data visualization.
- Learned to interpret analytical results and convert them into meaningful insights.
- Understood key factors influencing customer preferences beyond pricing.

Any Questions

You are Welcome!

THANK YOU

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